

CRISPR in action: practice and theory of
genome engineering

Introduction to CRISPR-Cas9

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Why CRISPR is so important and revolutionary?



PRECISION



SIMPLICITY



BROAD
APPLICATIONS



AFFORDABILITY



SPEED

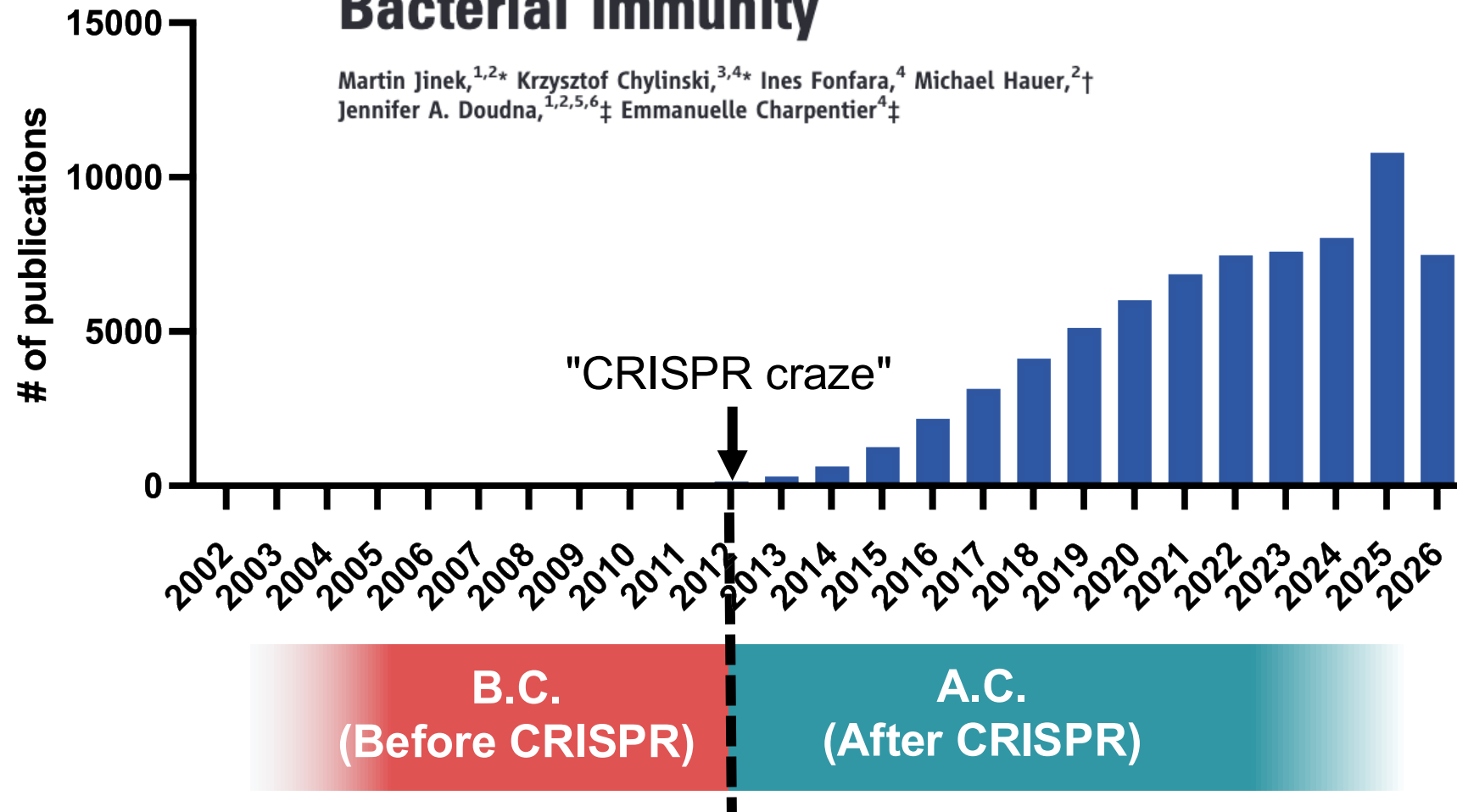


POTENTIAL

Annual number of publications with "CRISPR" in title or abstract (1970–2026)

A Programmable Dual-RNA–Guided DNA Endonuclease in Adaptive Bacterial Immunity

Martin Jinek,^{1,2*} Krzysztof Chylinski,^{3,4*} Ines Fonfara,⁴ Michael Hauer,^{2†}
Jennifer A. Doudna,^{1,2,5,6‡} Emmanuelle Charpentier^{4‡}



BC – Before CRISPR

- **Zinc Finger Nucleases (ZFNs):** Custom proteins engineered to bind and cut specific DNA sequences. Effective but complicated to design, expensive, and difficult to scale.

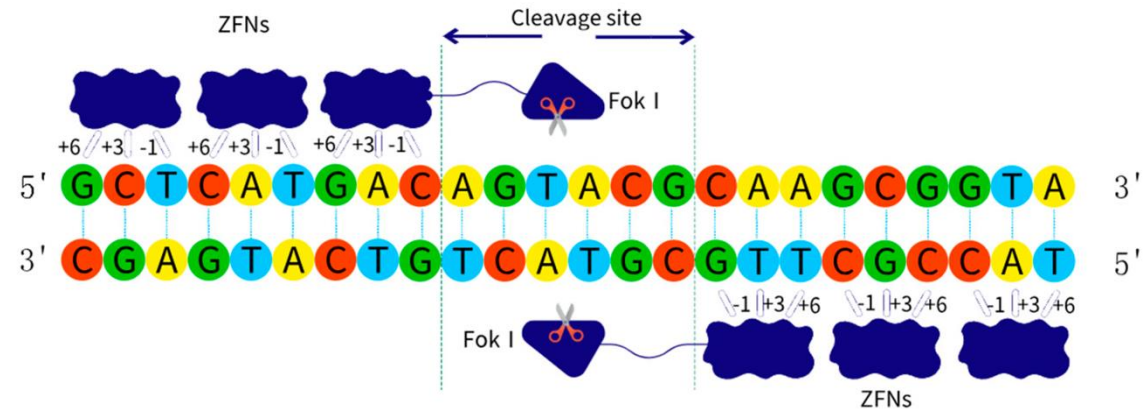


Figure Ye et al., 2022

- **TALENs (Transcription Activator-Like Effector Nucleases):** Modular proteins providing more flexibility and precision than ZFNs, but still labor-intensive and challenging to scale.
- **Homologous Recombination:** Used for gene targeting before programmable nucleases. It is precise but extremely inefficient and time-consuming for most organisms.

CRISPR (Clustered Regularly Interspaced Short Palindromic Repeats)



Nucleotide Sequence of the *iap* Gene, Responsible for Alkaline Phosphatase Isozyme Conversion in *Escherichia coli*, and Identification of the Gene Product

YOSHIZUMI ISHINO, HIDEO SHINAGAWA, KOZO MAKINO, MITSUKO AMEMURA, AND ATSUO NAKATA*
Department of Experimental Chemotherapy, The Research Institute for Microbial Diseases, Osaka University, 3-1 Yamadaoka, Suita, Osaka 565, Japan

Received 1 May 1987/Accepted 22 August 1987

An unusual structure was found in the 3'-end flanking region of *iap* (Fig. 5). Five highly homologous sequences of 29 nucleotides were arranged as direct repeats with 32 nucleotides as spacing. The first sequence was included in the putative transcriptional termination site and had less homology than the others. Well-conserved nucleotide sequences containing a dyad symmetry, named REP sequences, have been found in *E. coli* and *Salmonella typhimurium* (28) and may act to stabilize mRNA (18). A dyad symmetry with 14 nucleotide pairs was also found in the middle of these sequences (underlining, Fig. 5), but no homology was found between these sequences and the REP sequence. So far, no sequence homologous to these has been found elsewhere in procaryotes, and the biological significance of these sequences is not known.

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TGA AAATGGGAG G GAGTTC TACCGC AGAGGCGGGGGA ACTC CAAGT GATATCCATCATCGCATCCAGT GCGCC (1,451)
(1,452) CGGTTTATCCCCGCTGATGCGGGGAACACCAGCGTCAGGCGTGAAATCTACCGTCGTTGC (1,512)
(1,513) CGGTTTATCCCTGCTGGCGCGGGGA ACTC TCGGTT CAGGCGTTGCAAACCTGGCTACCGGG (1,573)
(1,574) CGGTTTATCCCCGCTAACGCGGGGA ACTC GTAGTCCATCATTCACCTATGTCTGA ACTCC (1,634)
(1,635) CGGTTTATCCCCGCTGGCGCGGGGA ACTCG (1,664)

consensus: CGGTTTATCCCCGCTGGCGCGGGGA ACTCAA
```

Figure 5.

Hypothesized their function as a
bacterial immune system
(Mojica et al., 2005)

First time the term
CRISPR was used
(Jansen et al., 2002)

Experimental proof that CRISPR
functions as a bacterial immune system
(Barrangou et al., 2007)

of publications

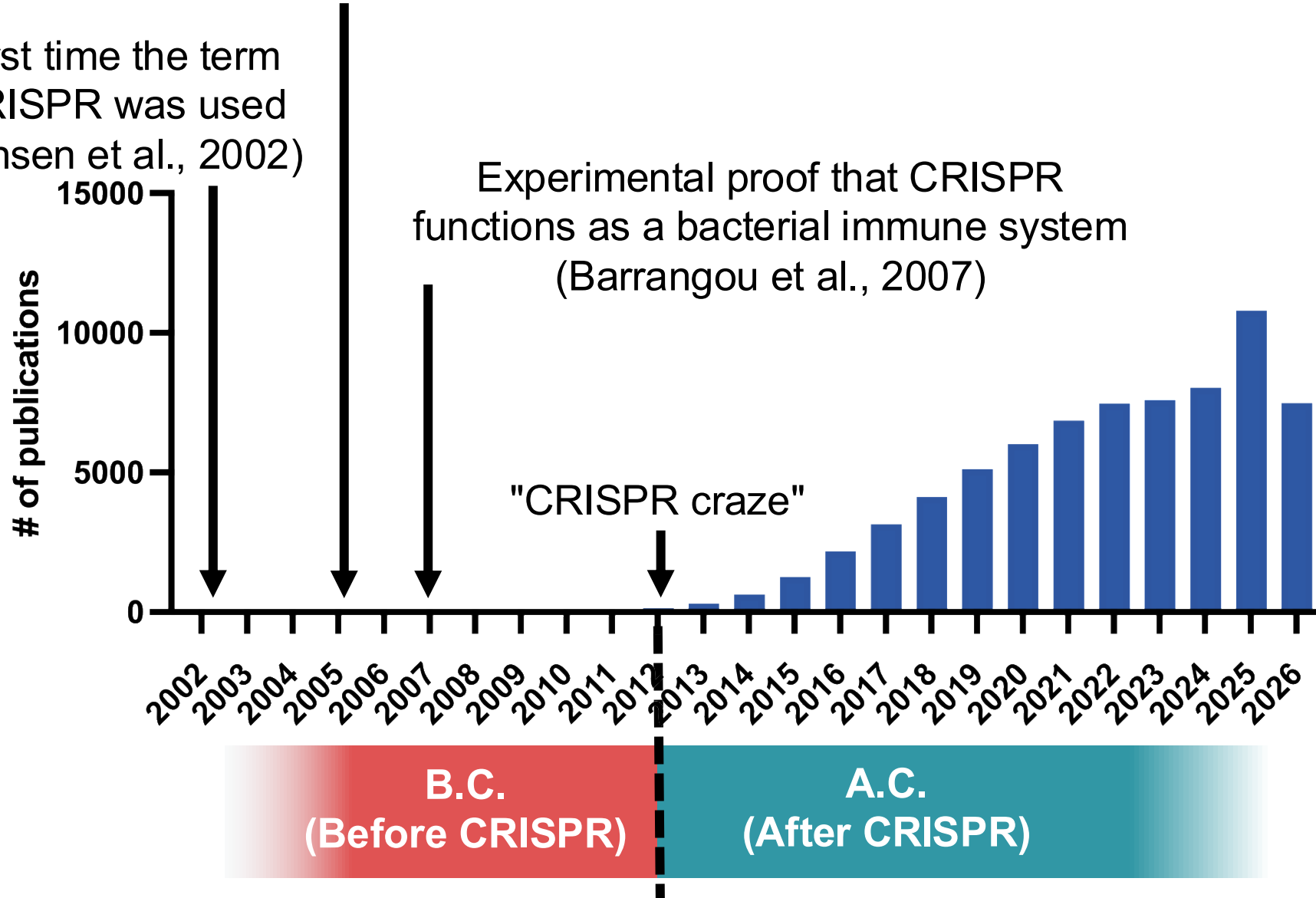
15000
10000
5000
0

"CRISPR craze"

2002 2003 2004 2005 2006 2007 2008 2009 2010 2011 2012 2013 2014 2015 2016 2017 2018 2019 2020 2021 2022 2023 2024 2025 2026

B.C.
(Before CRISPR)

A.C.
(After CRISPR)



CRISPR in Nature - Bacterial immune systems

Protect bacteria against foreign DNA elements, such as bacetriophages and cojugative elements

Innate immunity

- Restriction-modification (R-M) systems
- Abortive infection (Abi) systems

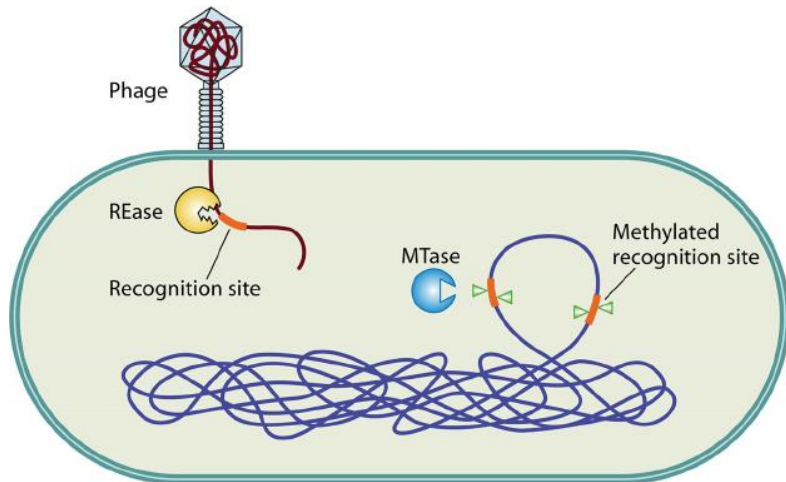


Figure Vasu et al., 2003

Adaptive immunity

- CRISPR-Cas systems

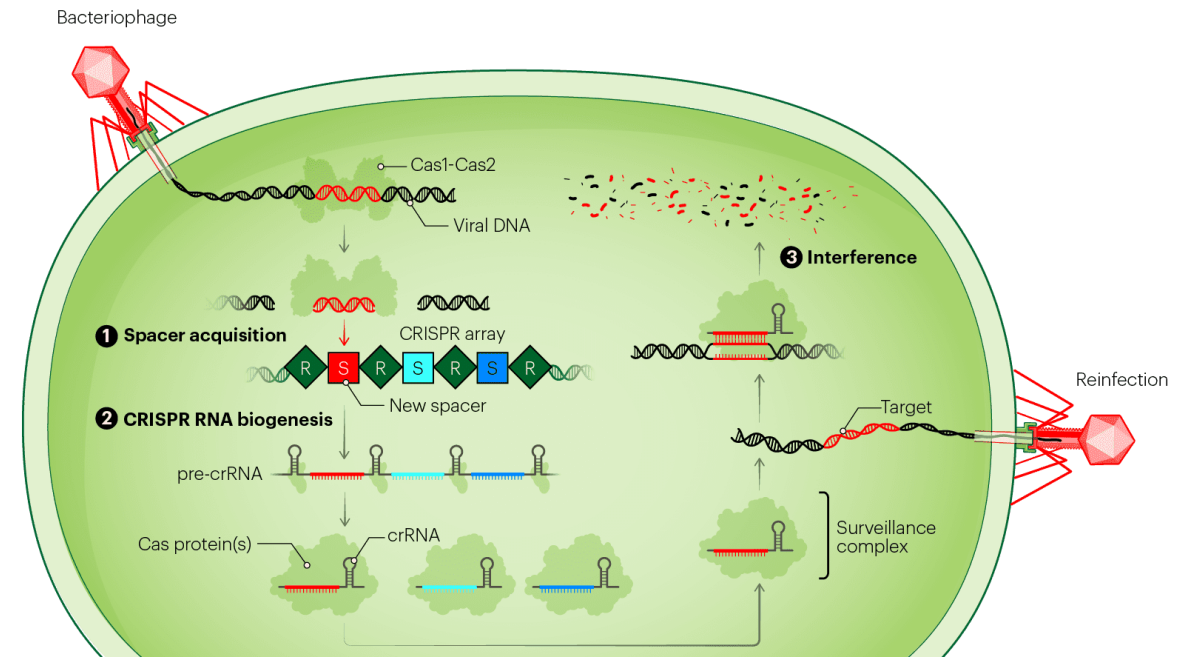
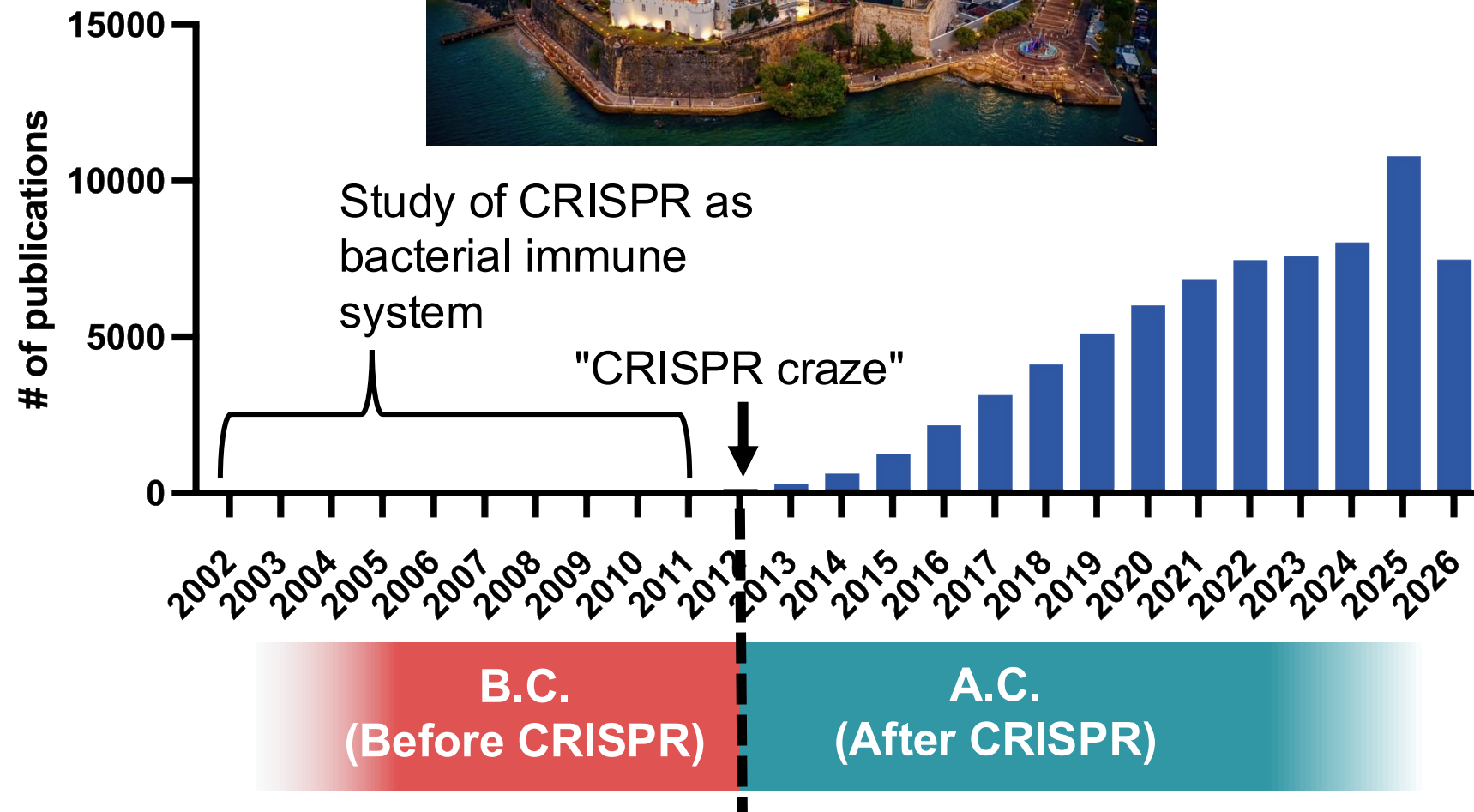


Figure CRISPRpedia-Innovative Genomics Institute

In 2011 Emmanuelle Charpentier discovered the tracrRNA in the CRISPR-Cas9 system of *Streptococcus pyogenes* (Deltcheva et al., 2011)

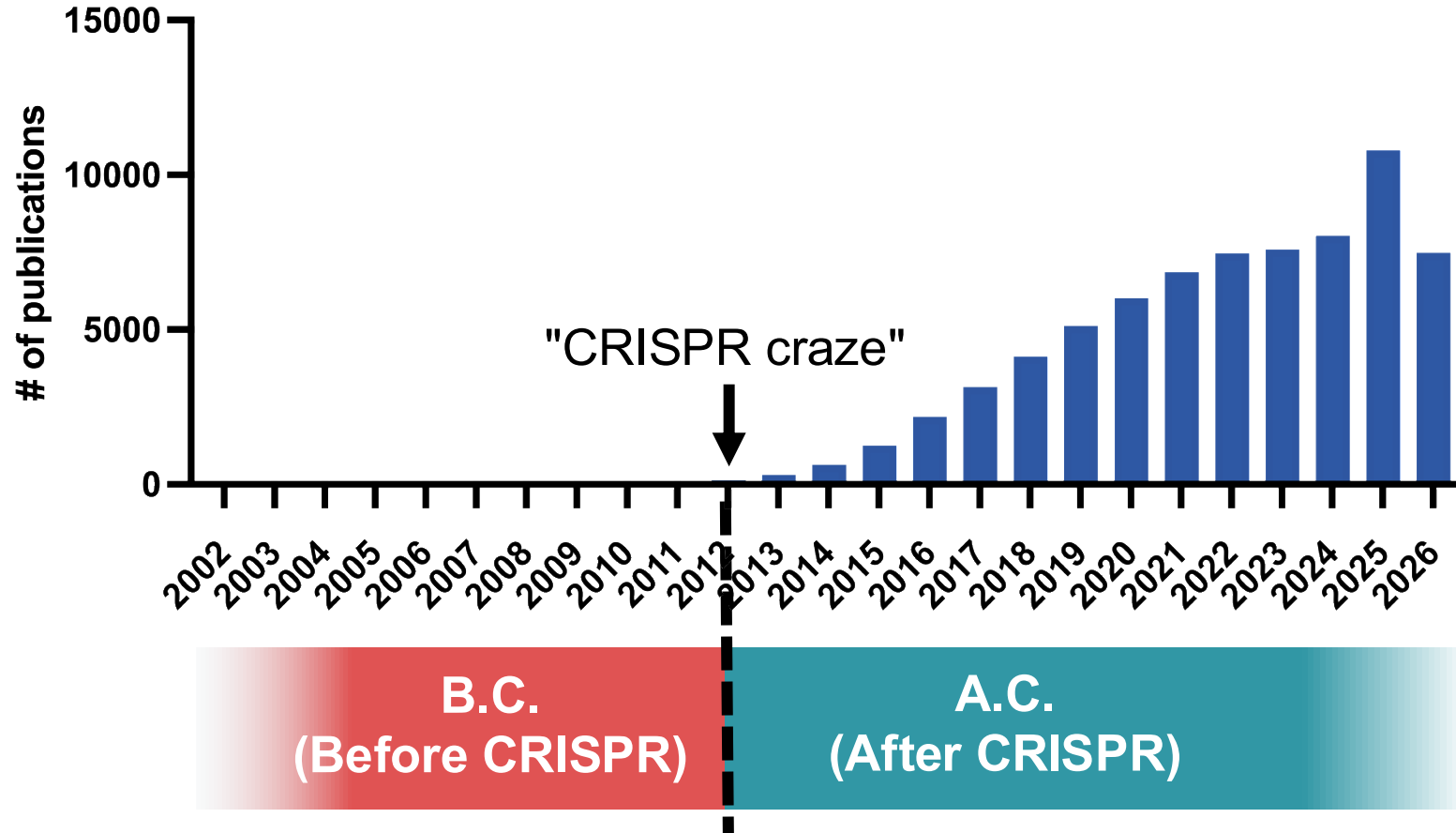


In 2011 Emmanuelle Charpentier and Jennifer Doudna meet in a conference in Puerto Rico and decide to collaborate



A Programmable Dual-RNA–Guided DNA Endonuclease in Adaptive Bacterial Immunity

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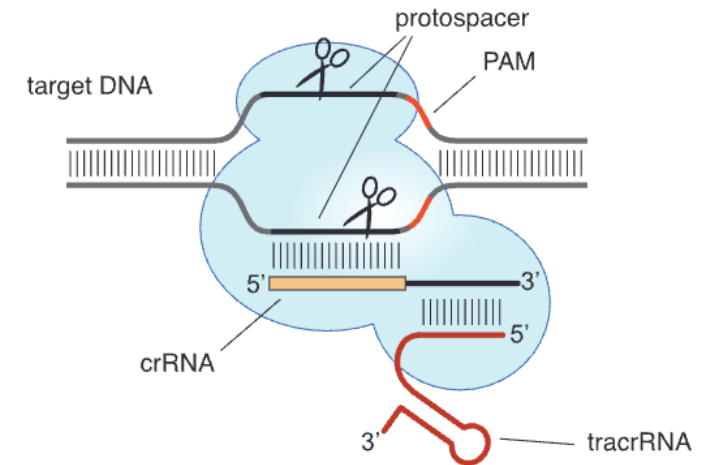
The group of **Virginijus Šikšnys** submitted their paper around the same time demonstrating CRISPR-Cas9 system could be programmed to cut DNA at specific sites in bacteria, but it was delayed in publication due to journal rejections, eventually appearing in **PNAS** in late **September 2012**, just **3 months** after the famous **Science** paper from Charpentier and Doudna.

A Programmable Dual-RNA–Guided DNA Endonuclease in Adaptive Bacterial Immunity

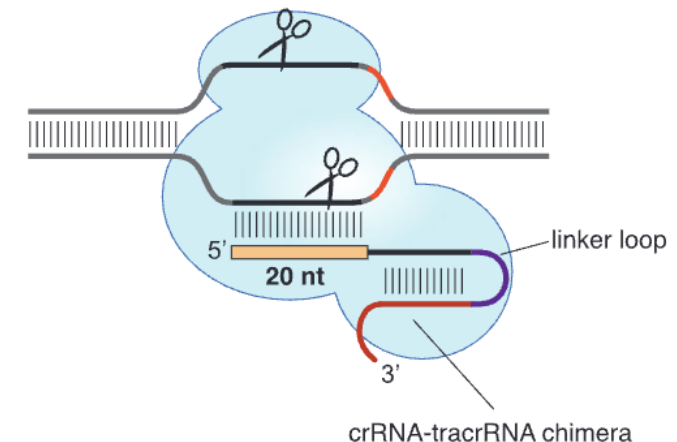
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- The paper described that CRISPR-Cas9 system are programmable and can be used for DNA-cutting in specific sites.
- *Streptococcus pyogenes* Cas9 = SpCas9
- SpCas9 has two active domains (HNH and RuvC-like) and each cleave one strand of the DNA
- They found that the SpCas9 system had some tolerance for single mismatches between the gRNA and the DNA target, but this tolerance was highly dependent on the position of the mismatch.
- They engineered a single chimeric RNA combining crRNA and tracrRNA, which can also program Cas9 for sequence-specific DNA cleavage.
 - chimeric crRNA + tracrRNA = gRNA = sgRNA

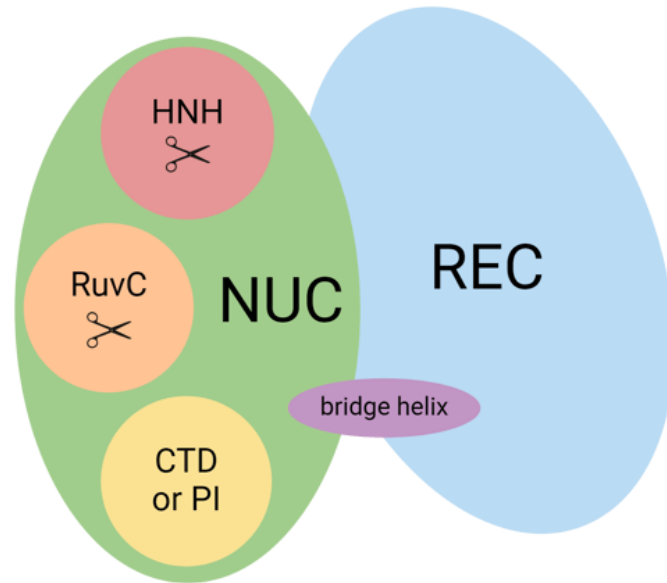
Cas9 programmed by crRNA:tracrRNA duplex



Cas9 programmed by single chimeric RNA

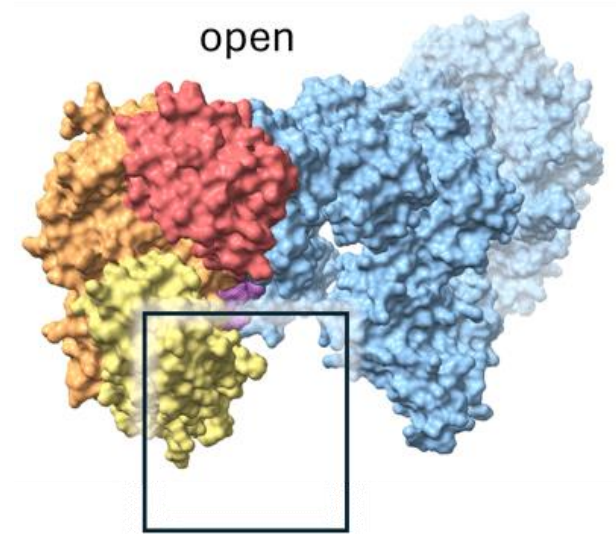


How CRISPR-Cas9 works?

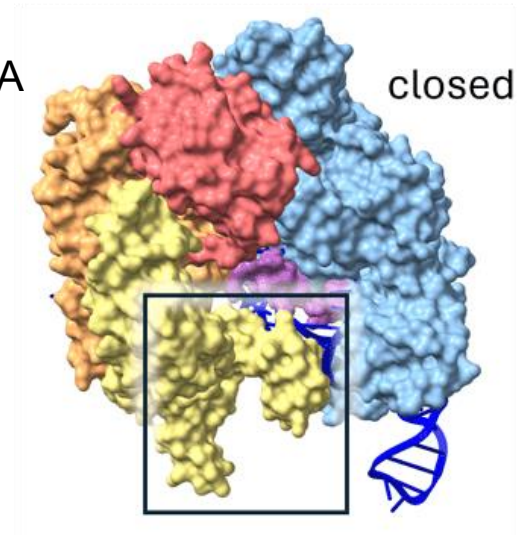


A cartoon depiction of apo Cas9's two major lobes, REC (recognition) and NUC (nuclease), and their subdomains. NUC includes the HNH and RuvC catalytic domains, as well as the CTD (also known as the PI = PAM interacting), while REC includes most of the bridge helix.

Apo SpCas9

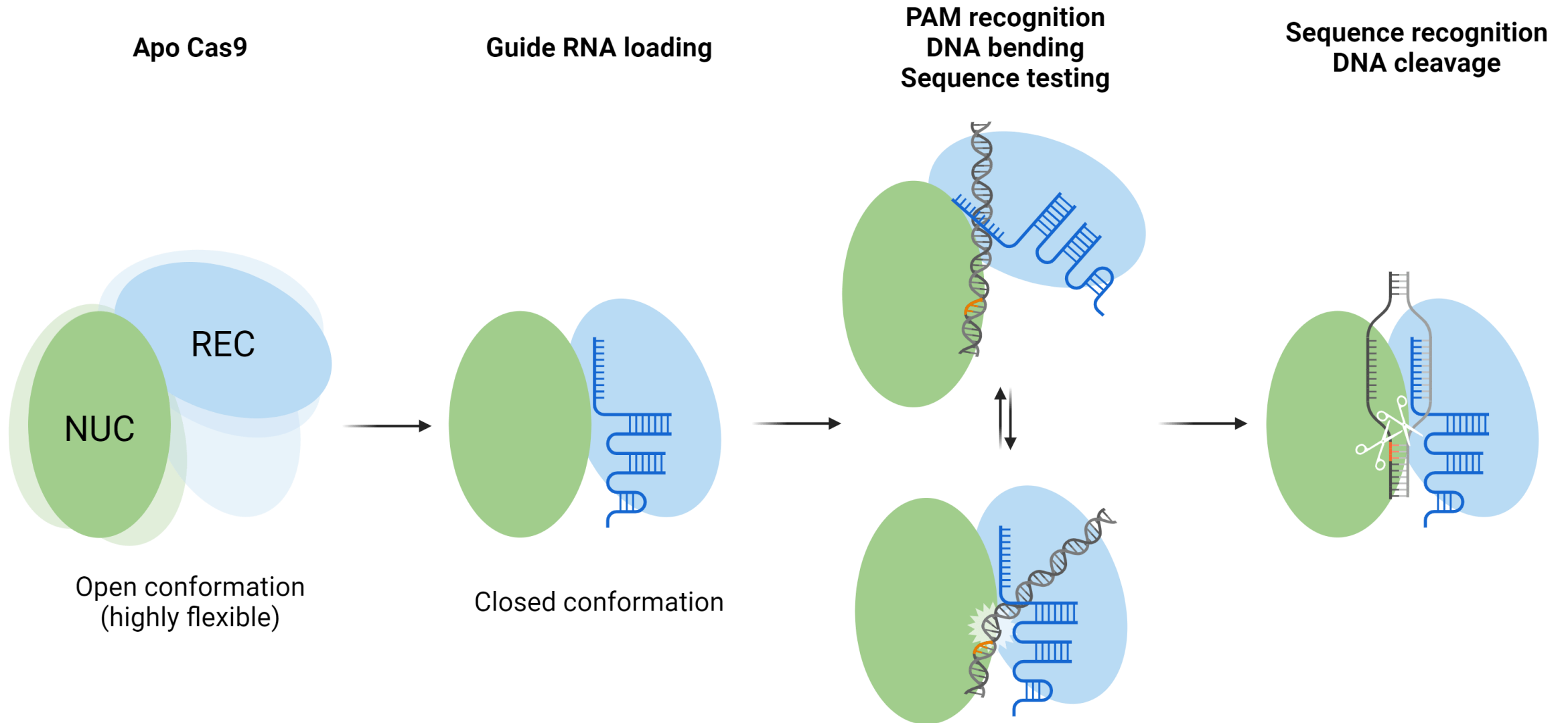


SpCas9:gRNA

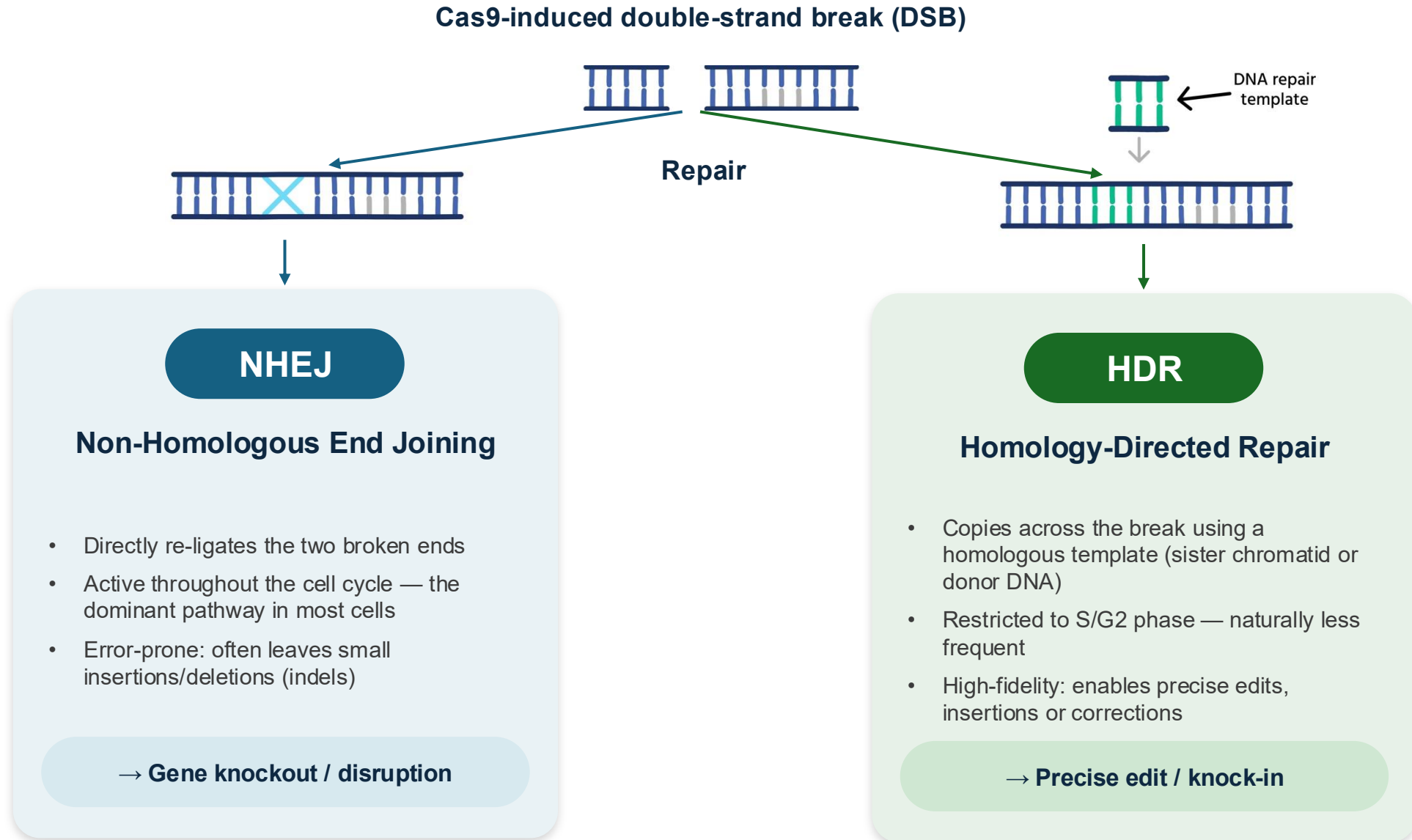


Crystal structure of SpCas9 in the apo state (PDB ID 4CMP) compared to the guide RNA-bound state (PDB ID 4ZT0)

How CRISPR-Cas9 works?

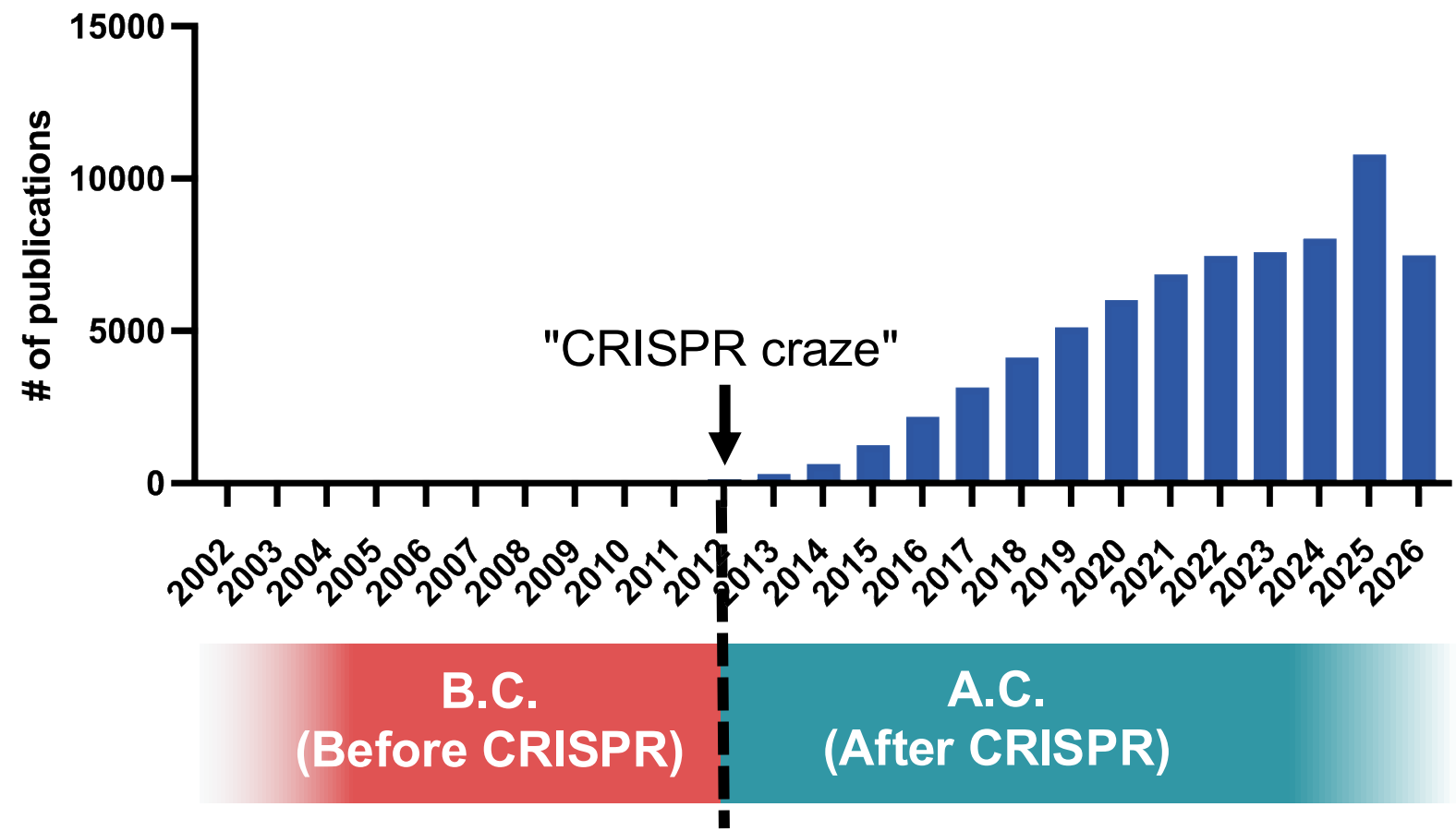


What happens after the cut? DNA repair pathways



Pathway choice can be biased experimentally — e.g., cell-cycle synchronization or small-molecule NHEJ inhibitors to favor HDR

AC – After CRISPR



AC – After CRISPR key advances

- **Explosion of applications:** CRISPR has been applied in bacteria and various eukaryotic organisms, including plants and humans. Uses now include multi-gene targeting, silencing, activation, antimicrobials, biosensors, and more. Many applications employ Cas proteins beyond the original effectors.
- **New CRISPR-Cas systems identified:** Before 2012, only three types were known. Three more have since been identified, including:
 - Type V: Cas12 family, single effector targeting DNA; some variants (like Cas14) also target single-stranded DNA.
 - Type VI: Cas13 proteins, which target RNA.
- **Discovery of anti-CRISPRs:** Anti-CRISPR proteins have been found to potently inhibit and regulate Cas activity.
- **Nobel Prize awarded:** Jennifer Doudna and Emmanuelle Charpentier received the Nobel Prize in Chemistry in 2020.
- **CRISPR-based medicines approved:** The first CRISPR-based therapies were approved in 2023 to treat sickle cell disease and transfusion-dependent beta thalassemia.
- **Ongoing patent dispute:** Legal battles over ownership of foundational CRISPR-Cas9 technology are still ongoing.

